

LAKE CUMBERLAND, KY, USA

WMO: 724354

Lat: 37.054N

Lon: 84.615W

Elev: 283

StdP: 97.98

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 63815

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.0	-17.2	0.9	-8.5	-14.0	1.2	-6.6	9.5	9.3	8.5	8.5	1.5	0	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	34.0	23.5	32.7	23.1	31.7	22.7	25.3	31.7	24.5	30.3	23.8	29.1	2.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.4	27.4	22.7	18.0	26.9	22.4	17.6	26.4	78.8	31.2	75.5	30.4	72.4	29.2	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.9	5.6	-14.2	35.4	4.2	2.7	-17.2	37.3	-19.7	38.9	-22.0	40.4	-25.0	42.3		
(5)			-15.2	26.0	4.1	1.1	-18.1	26.8	-20.5	27.4	-22.8	28.1	-25.8	28.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	2.8	4.9	8.7	14.3	19.0	23.2	24.9	24.4	21.1	14.7	8.4	4.4	(6)
(7)		DBStd	9.16	6.51	6.07	5.54	5.02	4.01	2.89	2.32	2.71	3.58	4.70	5.11	5.64	(7)
(8)		HDD10.0	801	237	163	92	18	1	0	0	0	0	12	89	189	(8)
(9)		HDD18.3	2218	481	378	299	138	43	3	0	1	16	130	298	431	(9)
(10)		CDD10.0	2362	15	19	52	146	279	396	461	445	332	159	42	17	(10)
(11)		CDD18.3	738	1	1	2	17	63	148	202	187	98	18	2	0	(11)
(12)		CDH23.3	6693	1	1	26	182	563	1338	1796	1688	903	185	11	1	(12)
(13)		CDH26.7	2579	0	0	2	33	155	518	745	734	351	41	1	0	(13)
(14)	Wind	WSAvg	1.9	2.4	2.4	2.5	2.4	1.8	1.5	1.3	1.2	1.2	1.5	1.9	2.1	(14)
(15)	Precipitation	PrecAvg	1259	119	100	128	101	125	118	114	96	77	79	85	120	(15)
(16)		PrecMax	1561	220	175	287	229	274	259	206	198	152	159	186	345	(16)
(17)		PrecMin	930	44	40	54	31	57	46	41	25	5	9	21	36	(17)
(18)		PrecStd	185	40	38	61	44	56	56	41	42	43	43	39	67	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	22.5	26.3	29.7	32.1	34.1	36.0	36.9	35.8	30.9	25.2	21.1	(19)
(20)			MCWB	15.6	14.7	15.9	19.3	21.6	24.1	24.6	23.4	22.2	20.7	16.4	15.7	(20)
(21)		2%	DB	17.6	19.8	23.0	27.4	30.1	32.7	33.4	34.0	32.6	27.5	22.3	17.7	(21)
(22)			MCWB	13.8	14.1	14.7	17.1	20.5	23.3	24.1	23.3	21.4	18.3	15.0	14.0	(22)
(23)		5%	DB	15.9	17.3	21.1	25.9	28.0	31.4	32.3	32.5	30.9	25.8	19.5	16.0	(23)
(24)			MCWB	12.7	11.8	13.8	16.2	19.7	22.6	23.7	23.2	21.2	17.4	14.1	12.7	(24)
(25)		10%	DB	12.7	14.8	17.9	23.0	27.1	29.9	31.0	31.1	28.6	22.8	17.3	13.0	(25)
(26)			MCWB	9.4	10.1	11.9	15.2	19.2	21.8	23.2	22.8	20.3	16.5	12.6	10.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	16.6	17.7	20.3	23.2	25.7	26.7	26.0	24.2	22.2	18.8	17.2	(27)
(28)			MCDWB	19.1	20.1	23.7	27.5	29.4	32.8	34.1	32.9	30.9	26.8	22.8	19.1	(28)
(29)		2%	WB	14.8	14.9	16.0	18.8	21.9	24.6	25.5	25.1	23.0	20.6	16.9	15.1	(29)
(30)			MCDWB	17.4	18.7	21.2	24.8	27.5	30.7	31.7	31.6	28.9	24.3	20.3	17.3	(30)
(31)		5%	WB	12.9	12.9	14.7	17.5	20.9	23.7	24.6	24.1	22.3	19.2	15.2	12.9	(31)
(32)			MCDWB	15.3	15.8	19.3	22.9	26.0	29.2	30.2	30.0	27.7	23.2	18.0	15.1	(32)
(33)		10%	WB	10.2	10.9	13.0	16.4	20.0	22.8	23.9	23.4	21.5	17.8	13.2	11.0	(33)
(34)			MCDWB	12.4	13.9	17.2	21.6	24.9	27.9	28.9	28.6	26.3	21.3	16.4	13.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.5	11.8	12.9	12.0	11.6	11.0	11.9	12.7	13.0	11.5	9.3	(35)
(36)			MCDBR	11.5	14.4	15.6	15.5	14.2	13.6	13.0	14.3	16.1	16.1	14.8	11.5	(36)
(37)			MCWBR	8.4	9.3	9.0	7.7	6.3	5.6	5.5	5.9	6.5	7.9	8.7	8.1	(37)
(38)		5% WB	MCDWB	10.3	11.5	12.5	12.3	11.8	11.5	11.0	12.0	12.2	12.0	10.9	10.0	(38)
(39)			MCWBR	8.1	8.9	8.1	6.9	5.6	5.4	5.1	5.3	5.4	6.6	8.0	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.319	0.352	0.396	0.448	0.465	0.496	0.499	0.414	0.361	0.323	0.305	(40)	
(41)		taud	2.495	2.457	2.407	2.281	2.167	2.156	2.089	2.066	2.287	2.428	2.499	2.518	(41)	
(42)		Ebn at Noon	884	917	913	885	841	824	794	781	836	855	859	864	(42)	
(43)		Edh at Noon	81	94	109	131	149	151	160	161	122	96	80	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.04	2.66	3.79	5.01	5.58	6.15	5.77	5.45	4.55	3.50	2.48	1.78	(44)	
(45)		RadStd	0.22	0.37	0.34	0.37	0.50	0.34	0.30	0.29	0.48	0.45	0.27	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	+85

Nomenclature: See separate page